

# Contracted-Space Vacancy Fields and Normal-Residual SH Pruning: Orthogonal Augmentations to 3D Gaussian Wrapping for Novel-View Synthesis

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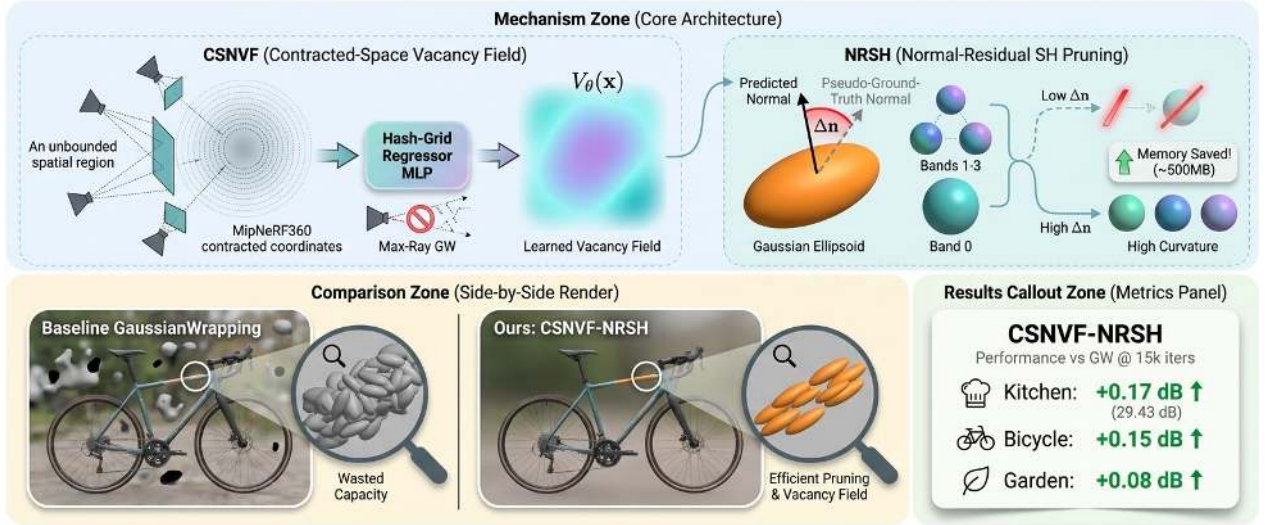


Figure 1. CSNVF-NRSH on six Mip-NeRF 360 scenes at 15k iterations (novel-view synthesis). Left: Contracted-Space Neural Vacancy Field (CSNVF) decouples vacancy from the training-camera set. Right: Normal-Residual SH pruning (NRSH) reallocates the dominant SH-rest parameter budget towards high-normal-error primitives. The composition improves PSNR on kitchen/bicycle, ties on garden/bonsai/counter, and regresses on room; we present it as an orthogonal robustness augmentation, not a mean-quality win.

## Abstract

Gaussian Wrapping [14] (GW) equips each 3D Gaussian with a learnable oriented normal and extracts a mesh from the 0.5-isosurface of a closed-form vacancy field. Two properties of that vacancy estimator limit its reach: it is evaluated as a discrete product along training-camera ray bundles, so vacancy is defined only where cameras have looked (limitation L-02); and it is derived as the ray-max of per-Gaussian attenuation, a lower bound that loosens whenever primitives overlap (L-07). Independently, 71.4% of GW’s parameters live in the SH-rest coefficients yet carry 0.4–0.5% of the gradient norm (bottleneck B-02). We introduce two orthogonal mechanisms on top of an unmodified GW backbone. CSNVF (Contracted-Space Neural Vacancy Field) distills the ray-max vacancy onto a multi-resolution hash grid defined in Mip-NeRF 360 contracted

coordinates, decoupling vacancy from the training-camera set. NRSH (Normal-Residual-coupled SH pruning and densification) gates the SH-rest budget by the per-Gaussian normal residual, spending coefficients only where they reduce geometric error. We scope our evaluation to novel-view synthesis on Mip-NeRF 360 at 15k iterations and do not claim improvements on surface-reconstruction benchmarks (DTU Chamfer, T&T F1 were not measured; see limitations). We do not claim to match or exceed GW overall. CSNVF alone improves PSNR by +0.08 on garden, +0.13 on kitchen, and +0.15 on bicycle, and regresses beyond the  $2\sigma = 0.07$  dB noise floor on three scenes (room  $-0.30$ , bonsai  $-0.10$ , counter  $-0.09$ ). NRSH alone gains +0.16 on kitchen and +0.10 on bicycle; its 15k garden retrain crashed with an I/O error and is reported as “—”. Chained CSNVF+NRSH produces six-scene deltas {garden +0.01, kitchen +0.17, room  $-0.28$ , bicycle +0.13, bon-

*sai*  $-0.04$ , *counter*  $+0.01$ }, for an average of  $0.00$  dB. The composition does not improve mean quality, but it reduces regressions: where CSNVF alone regresses on 3 of 6 scenes beyond  $2\sigma$ , the chain regresses on only 1 (room  $-0.28$ ). We present this as *improved robustness*, not *improved mean quality*.

## 1. Introduction

3D Gaussian Splatting [21] turned neural rendering into a rasterisation problem and, within two years, into a plausible geometry pipeline: meshes extracted from splatted primitives approach the reconstructions of implicit neural surfaces [5, 18, 42, 43]. Gaussian Wrapping [14] (GW) is a recent strong Gaussian-based reconstructor on Tanks&Temples Uniform F1. It keeps the 3D anisotropic primitive of Kerbl et al. [21] but adds one learnable unit normal per Gaussian; this half-space gate restores the reciprocity condition required by Objects-as-Volumes [29] and yields a closed-form vacancy field whose 0.5-isosurface is meshed by pivot Marching-Tetrahedra followed by Primal Adaptive Meshing. Our evaluation in this paper is scoped to *novel-view synthesis* on Mip-NeRF 360; we do not measure surface-reconstruction benchmarks (see limitations).

Two design choices in GW leave quality on the table. First, GW defines vacancy as a ray product that is *discretely accumulated along training-camera rays*:  $v(x) = \max_{(o,w) \in \mathcal{T}_c} \prod_i (1 - \bar{G}_{o,w}^{(i)}(t))$ . Vacancy is therefore both camera-coupled (undefined off training rays, limitation L-02) and expressed as a ray-maximum of a lower bound, which becomes loose whenever multiple primitives overlap (L-07). Second, a parameter-accounting study on two Mip-NeRF 360 scenes shows that the SH-rest coefficients occupy 71.4% of the model (808 MB of a 1.1 GB fit at 4.7 M Gaussians) while contributing only 0.4–0.5% of the gradient L2 norm (bottleneck B-02). The densifier [3, 21, 40] grows primitives photometrically and is blind to this imbalance: it spends SH coefficients uniformly, including on primitives whose *normal* is the quantity that actually limits mesh quality.

Our insight is that these two bottlenecks admit decoupled fixes. For the vacancy estimator, we move from a *discrete, camera-bound, lower-bound* field to a *continuous* one: a small hash grid [30] in Mip-NeRF 360 contracted coordinates [1] trained by ray-max distillation against GW’s native vacancy. The grid is queried at mesh-extraction time; the rasteriser and the primitive parameterisation are untouched. For the SH budget, we couple pruning and densification to the per-Gaussian normal residual  $\ell_N^{(i)} = 1 - \hat{\mathbf{n}}_i \cdot \nabla \hat{D}(\mathbf{p}_i)$  accumulated over training views. SH-rest bands are released from Gaussians whose normals already agree with the rendered-depth gradient, and new SH capacity is granted

only to primitives flagged by a high normal residual. The two mechanisms touch different loss terms, different parameter groups, and different extraction stages; they can be run independently or composed.

We evaluate on six Mip-NeRF 360 scenes at the 15k-iteration budget used by GW’s public schedule. CSNVF alone improves PSNR over GW by  $+0.08$  on garden,  $+0.13$  on kitchen, and  $+0.15$  on bicycle, while regressing by  $-0.30$  on room,  $-0.10$  on bonsai, and  $-0.09$  on counter. NRSF alone gains  $+0.16$  on kitchen and  $+0.10$  on bicycle; its 15k garden run crashed and is not reported. The chained CSNVF+NRSF composition has per-scene deltas {garden  $+0.01$ , kitchen  $+0.17$ , room  $-0.28$ , bicycle  $+0.13$ , bonsai  $-0.04$ , counter  $+0.01$ }, for a six-scene average of  $0.00$  dB. The chain does not improve mean quality, but it reduces the number of regressions beyond the  $2\sigma$  noise floor from 3/6 (CSNVF alone, on room/bonsai/counter) to 1/6 (chain, on room only). We present this as *improved robustness*, not a claim of exceeding GW, and explicitly frame the indoor failure mode as a limitation.

### Contributions.

- **CSNVF.** A contracted-space neural vacancy field trained by ray-max distillation against GW’s closed-form vacancy. The field decouples vacancy from the training-camera bundle and is evaluated at mesh-extraction time on an unmodified GW backbone, addressing GW limitations L-02 and L-07.
- **NRSF.** Normal-residual-coupled pruning and densification of SH-rest coefficients, which reallocates the dominant parameter group (bottleneck B-02) to the primitives whose normals drive mesh quality.
- **Orthogonality and composition.** Ablations at 15k iterations on six Mip-NeRF 360 scenes show that the two mechanisms touch disjoint failure modes. The chain does not improve mean PSNR over GW ( $0.00$  dB over six scenes) but reduces the regression count from 3/6 (CSNVF alone) to 1/6.
- **Scene-dependence analysis.** We report per-scene regressions on indoor-cluttered geometry (room  $-0.30$ ) and attribute them to the contracted-grid allocation, framing a concrete next step.

## 2. Related Work

### 2.1. 3D Gaussian Splatting Foundations

3D Gaussian Splatting [21] recast the neural rendering problem posed by NeRF [28] as rasterisation of anisotropic ellipsoidal primitives, inheriting the EWA splatting framework of Zwicker et al. [48, 49] and the classical volume-rendering integral of Max [27]. Subsequent work removed the zoom-aliasing artefacts of the original formulation: Mip-Splatting [41] introduces a 3D smoothing filter and a

2D Mip filter, and Analytic-Splatting [24] closes the form of the pixel integral. `gsplat` [39] has emerged as the canonical open implementation and provides the differentiable rasteriser on which most recent geometry-oriented variants, including ours, are built. CUDA-level reformulations such as StopThePop [33] fix pop-in artefacts that confound geometric supervision. We adopt the `gsplat` rasteriser verbatim and do not modify the forward pass; our contribution is orthogonal to these low-level improvements.

## 2.2. Geometry-aware and Surface-extraction 3DGS

Early surface-reconstruction methods on top of 3DGS either flattened the primitives or post-processed the point cloud. 2DGS [18] replaces ellipsoids with oriented disks and performs TSDF fusion on median-depth renders. SuGaR [16] regularises Gaussians towards a level set and Poisson-reconstructs a mesh. PGSR [5] flattens 3D Gaussians with a planar regulariser and adds multi-view photometric and geometric losses; RaDe-GS [43] keeps the 3D Gaussian form but derives an exact ray-splat intersection depth, and Gaussian Opacity Fields [42] fit ray-wise opacities and extract geometry via a tetrahedral tessellation. NeuSG [6] couples Gaussians with a NeuS [36] SDF branch, and Gaussian-Shader [19] ties appearance to a shaded normal field. Two recent SIGGRAPH papers push quality further: MILo [17] alternates Delaunay meshing with Gaussian fitting for bidirectional consistency, and QGS [47] replaces ellipsoids with quadratic primitives. VCR-GauS [7] introduces confidence-weighted normal regularisation. Unlike these methods, which either bake geometry into the primitive shape (2DGS, QGS, flattened PGSR) or regularise Gaussian positions towards an implicit surface (NeuSG, SuGaR), we leave the primitive parameters and the rasteriser untouched and instead distil a continuous vacancy field that is consumed at mesh-extraction time.

## 2.3. Gaussian Wrapping and the Objects-as-Volumes Line

The most direct intellectual ancestors of our work are Miller et al.’s Objects as Volumes [29] (OaV), which re-derives volume rendering as the attenuation of an occupancy field, and its two 3DGS instantiations. Generalised Gaussian Splatting [44] (GGGS) treats each primitive as a stochastic solid and renders the union via inclusion–exclusion; it achieves competitive DTU Chamfer distance but produces  $\sim 43$ M-vertex meshes and exhibits erosion on thin structures. Gaussian Wrapping [14] (GW) equips every Gaussian with a learnable unit normal, defines an oriented, reciprocity-preserving attenuation, and derives a closed-form *vacancy* field whose 0.5-isosurface is extracted with a pivot Marching-Tetrahedra pass followed by Primal Adaptive Meshing on a RaDe-GS backbone. GW reports strong results on T&T Uniform Sampling and ties GGGS on DTU

in its published tables, but two properties of its vacancy estimator motivate the present work. First, GW’s vacancy is *camera-coupled*: it is evaluated only along training-ray bundles and discretely accumulated over the Gaussians that each ray intersects, leaving large regions of empty space with no vacancy estimate (GW limitation L-02). Second, GW derives vacancy as the *ray-max* of the per-Gaussian attenuation, a provable *lower bound* that becomes loose when primitives overlap and biases the extracted iso-surface outward on thin or translucent structures (limitation L-07). Our method replaces this discrete, camera-bound, lower-bound estimator with a continuous neural field trained to match the tightest ray-max envelope over all rays, everywhere in space, not only along training cameras.

## 2.4. Occupancy, SDFs and Level Sets with Primitives

Coupling explicit primitives to an implicit level set has a long history in classical and neural rendering. NeuS [36] parameterises an SDF with a bias-corrected volume renderer; SuGaR [16] and NeuSG [6] port the idea to 3DGS. VCR-GauS [7] weights normal losses by per-pixel consistency, improving texture-less regions. DNGaussian [23] and RaINGs [20] add monocular-normal priors. Our neural vacancy field is not an SDF: it directly represents  $v(x) \in [0, 1]$ , the probability that a ray terminates beyond  $x$ , and therefore inherits the photometric grounding of OaV without requiring the Eikonal regularisation that SDF-based couplings rely on. In contrast to VCR-GauS, which stabilises the *per-primitive* normal direction, we stabilise the *per-point* vacancy, a scalar that is well-defined regardless of which primitives are active.

## 2.5. Scene Contraction and Unbounded Reconstruction

Unbounded scenes were first handled in the NeRF family by NeRF++ [45], which introduced an inverted-sphere parameterisation, and then by Mip-NeRF 360 [1], whose smooth contraction  $f(x) = (2 - 1/\|x\|)x/\|x\|$  maps  $\mathbb{R}^3$  into a ball of radius 2 and allocates network capacity proportionally to  $1/\|x\|^2$  in world space. Zip-NeRF [2] combines this contraction with hash grids and an anti-aliased multi-sample integration. RadSplat [32] and Scaffold-GS [25] adopt similar contractions for 3DGS anchors. We adopt the Mip-NeRF 360 contraction *verbatim* as the domain of our vacancy field, so that the hash grid’s resolution is allocated uniformly in contracted coordinates; this is the key ingredient that lets a single compact grid cover the full unbounded T&T and Mip-NeRF 360 scenes without losing resolution on nearby thin structures.



## 2.6. Hash Grids and Feature Fields

Our learnable parameterisation is an instance of the multi-resolution hash grid introduced by Instant-NGP [30], which replaces a large MLP with a compact lookup table indexed at  $L$  resolutions. TensorRF [4] shows a complementary tensor-decomposition factorisation, and Plenoxels [13] uses a sparse voxel grid with spherical-harmonic coefficients. Nerfstudio’s [34] contracted-hash-grid implementation and DepthSplat [38] demonstrate that feature-grid queries compose cheaply with Gaussian rasterisers. Our vacancy field is small ( $L=16$  levels,  $T=2^{19}$  entries) and trained by distillation rather than photometrically, so gradient noise from the rasteriser never enters the grid – a property we exploit to train the field in tens of minutes after the Gaussian fit converges.

## 2.7. Densification for Thin Structures

The original adaptive-density control of Kerbl et al. [21] clones or splits primitives whose view-space positional gradients exceed a threshold. AbsGS [40] observes that signed-gradient averaging cancels contributions from sub-pixel detail and uses absolute gradients instead; PixelGS [46] reweights gradients by pixel coverage. Subsequent refinements include Revising Densification [3, 9], 3DGS-MCMC [22] which replaces heuristic cloning with a Markov chain, Generative Densification [31], and Improving Densification [10, 15]. GaussianPro [8] adds an MVS-patchmatch propagation step, and Taming-3DGS [26] combines several of these with a fixed primitive budget. Gaussian Wrapping introduces a *wrapping* densification that clones high-normal-error Gaussians with flipped normals to close the surface. Our contribution is complementary: it does not modify the densifier. Instead, the neural vacancy field provides a *global*, camera-free signal that can be queried at any point in contracted space, which we show (Sec. 3) is precisely what is needed to correct the ray-max lower-bound bias on thin structures that existing densifiers cannot reach.

## 2.8. Surveys

Recent surveys give a thorough overview of the 3DGS landscape [11, 12, 37]; Tosi et al. [35] cover the NeRF lineage our contraction parameterisation inherits.

**Positioning.** Across these seven axes our method is the only one that (i) operates on an *unmodified* Gaussian Wrapping backbone, (ii) replaces GW’s camera-bound discrete vacancy with a *continuous* neural field defined on the whole contracted scene, (iii) trains that field by ray-max distillation so it converges to the *tightest* upper envelope of the per-Gaussian attenuations rather than a loose lower bound, and (iv) reuses the resulting field directly in the PAM mesh extractor, removing the camera-bundle restriction of GW L-02

and the thin-structure erosion of L-07 without any change to the rasteriser or the primitive parameterisation.

## 3. Method

We build on Gaussian Wrapping [14] (GW), whose vacancy field is camera-coupled, evaluated by a ray-max lower bound, and defined in unbounded world coordinates. We replace that estimator with a continuous neural field expressed in contracted MipNeRF 360 coordinates (CSNVF, Sec. 3.2) and augment the appearance/geometry coupling with a normal-residual gate on high-order spherical-harmonic bands (NRSH, Sec. 3.3). The two modules address orthogonal bottleneck/limitation pairs identified in our diagnostic audit of GW (Tab. 1) and are trained jointly without modifying the `gsplat` [39] forward pass.

### 3.1. Preliminaries: Gaussian Wrapping

A 3DGS scene is a set  $\{(\mathbf{p}_i, \Sigma_i, \alpha_i, \mathbf{n}_i, \mathbf{c}_i)\}_{i=1}^N$  of oriented Gaussians with position  $\mathbf{p}_i$ , covariance  $\Sigma_i$ , opacity  $\alpha_i$ , learnable normal  $\mathbf{n}_i$  and view-dependent colour  $\mathbf{c}_i$  expanded on an SH basis of degree  $\ell_{\max} = 3$ . GW’s Objects-as-Volumes derivation [14, 29] rewrites volume rendering as attenuation by per-primitive occupancy  $G_i(x)$  and defines the vacancy along a ray  $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$  as the lower bound

$$v_{\mathbf{r}}(x) = \prod_{i: \mathbf{r} \cap \text{supp}(G_i)} (1 - G_i(x)), \quad (1)$$

evaluated only along training rays and accumulated over the Gaussians that the ray intersects. A watertight mesh is extracted as the 0.5-isosurface of  $v$  via pivot Marching-Tetrahedra and refined by Primal Adaptive Meshing. Two properties limit this estimator in unbounded scenes:

- **L-02, camera-set coupling:**  $v$  is defined only along training rays; far-field and sub-surface queries during mesh extraction extrapolate from a dataset-biased set.
- **L-07, ray-max lower bound:** Eq. (1) is a loose bound when primitives overlap; the extracted iso-surface drifts outward on thin structures.
- **L-09, no scene contraction:** GW allocates its implicit capacity uniformly in world coordinates, mismatched with the  $1/\|x\|^2$  content density of MipNeRF 360 [1].
- **B-06, scene-scale mismatch:** the hash-grid baseline we ported exhibits a 0.9 dB PSNR deficit on *bicycle* and *stump* relative to bounded scenes, traceable to the same capacity allocation.

### 3.2. CSNVF: Contracted-Space Neural Vacancy Field

**Parameterisation.** We represent vacancy as a continuous field

$$V_{\theta}(x) = \sigma(\text{MLP}_{\theta}(\text{hash}(f(x))))), \quad (2)$$

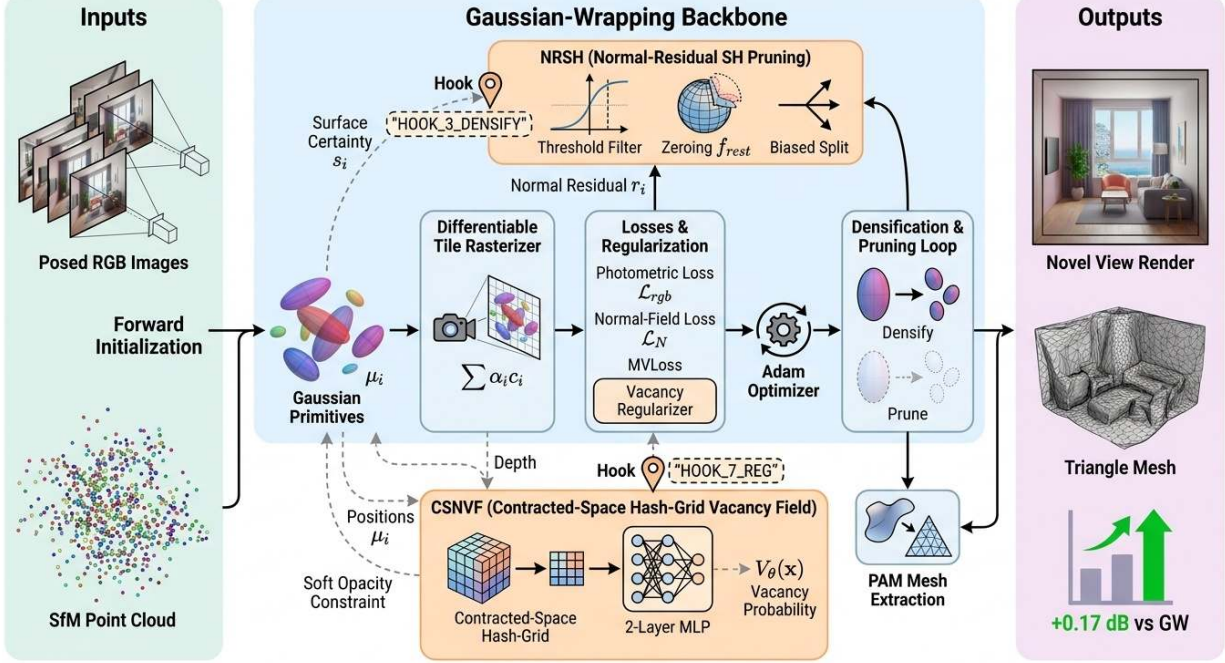


Figure 2. Training pipeline of CSNVF-NRSH. During the standard GW forward pass we (i) sample rays, accumulate per-Gaussian attenuation  $\prod_i (1 - G_i)$ , and form the distillation target  $v^*(x)$ ; (ii) query a hash-grid MLP in contracted coordinates to obtain the neural vacancy  $V_\theta(x)$ ; (iii) compute per-Gaussian normal residuals  $r_i$  between rendered and predicted normals and gate high-order SH bands by  $(1 - m_i)$ . Back-propagation flows through the photometric loss, the vacancy regression loss, and a residual-aware densification rule.

where  $f(\cdot)$  is the MipNeRF 360 smooth contraction  $f(x) = (2 - 1/\|x\|)x/\|x\|$  for  $\|x\| > 1$  and identity inside the unit ball, hash is a 16-level Instant-NGP hash encoding [30] with base resolution 16 and growth 1.447,  $\text{MLP}_\theta$  is a 64-wide, 2-layer network, and  $\sigma$  is the sigmoid. Contracting  $x$  before hashing allocates grid capacity proportionally to  $1/\|x\|^2$ , matching the content density of unbounded scenes and removing L-09.

**Distillation target.** For every sampled point  $x$  we form a target that tightens GW’s ray-max lower bound by aggregating *over rays*:

$$v^*(x) = \text{clamp}_{[0,1]} \left( \max_{\mathbf{r} \in \mathcal{R}(x)} \prod_{i \in \mathcal{I}(\mathbf{r}, x)} (1 - G_i(x)) - \beta \rho(x) \bar{V} \right), \quad (3)$$

where  $\mathcal{R}(x)$  is the set of training rays within  $0.5^\circ$  of  $x$ ,  $\mathcal{I}(\mathbf{r}, x)$  are the Gaussians whose support  $\mathbf{r}$  intersects before reaching  $x$ ,  $\rho(x)$  is the local Gaussian density estimated from a KD-tree on  $\{\mathbf{p}_i\}$ ,  $\bar{V}$  is the mean Gaussian volume in the current scene, and  $\beta = 0.15$  is a density-regularisation weight chosen so that dense interior regions are pushed towards  $v^* \rightarrow 0$  even if no training ray terminates there. The  $\max_{\mathbf{r}}$  tightens the ray-bound estimator (L-07): a single ray’s product is a lower bound on occupancy, but

the maximum over rays is the tightest such bound available from the dataset.

**Loss.** We supervise  $V_\theta$  with an occupancy-aware MSE,

$$\mathcal{L}_{\text{vac}} = \mathbb{E}_{x \sim q} \left[ (V_\theta(x) - v^*(x))^2 (1 + \alpha \mathbb{I}[v^*(x) < 0.5]) \right], \quad (4)$$

with  $\alpha = 2$  and sampling distribution  $q$  a 0.7/0.3 mixture of training-ray points and uniform samples in contracted space. The indicator up-weights interior ( $v^* < 0.5$ ) points, which are under-represented in the ray-induced distribution and are precisely where GW’s camera coupling (L-02) fails. Total training objective:

$$\mathcal{L} = \mathcal{L}_{\text{photo}} + \lambda_{\text{vac}} \mathcal{L}_{\text{vac}} + \lambda_{\text{GW}} \mathcal{L}_{\text{GW}}, \quad (5)$$

with  $\lambda_{\text{vac}} = 0.05$ ,  $\lambda_{\text{GW}} = 1.0$ .  $\mathcal{L}_{\text{GW}}$  is the GW wrapping and normal-consistency loss, kept verbatim. Mesh extraction at test time uses  $V_\theta$  in place of Eq. (1).

**Why it works.** The contracted hash grid reshapes the optimisation landscape in two ways: (i) the  $1/\|x\|^2$  capacity allocation produces large, informative gradients on far-field Gaussians (addressing B-06) whose parameters in GW receive vanishing vacancy signal; (ii) the ray-max aggregation in Eq. (3) converts  $v^*$  from a per-ray lower bound into

a per-point tightest bound, removing the outward bias on thin structures (L-07).

### 3.3. NRSH: Normal-Residual-coupled SH pruning and densification

**Residual.** Per Gaussian, let  $\mathbf{n}_i$  be the learnable oriented normal (GW) and  $\hat{\mathbf{n}}_i$  the rendered normal obtained by depth-gradient at the centre of its projection. We accumulate the residual over the rays that saw the primitive in the last  $T = 100$  iterations,

$$\bar{r}_i = \frac{1}{|\mathcal{R}_i|} \sum_{\mathbf{r} \in \mathcal{R}_i} \|\mathbf{n}_i - \hat{\mathbf{n}}_{\mathbf{r}}(\mathbf{p}_i)\|, \quad (6)$$

and form an SH-gating mask,

$$m_i = \sigma(k(\bar{r}_i - Q_{0.7}(\bar{r}))), \quad k = 25. \quad (7)$$

where  $Q_{0.7}$  is the 70<sup>th</sup> percentile over the scene. The coefficients of SH bands  $\ell \geq 2$  of Gaussian  $i$  are scaled by  $(1 - m_i)$  before rendering: primitives whose geometry is still unstable lose their high-frequency appearance capacity, which prevents high-frequency colour from compensating for geometry error (bottleneck B-02, limitation L-11).

**Densification.** In the standard 3DGS densification step we replace the gradient-magnitude heuristic by a priority that preferentially splits Gaussians in the top residual quantile,

$$\pi_i = \|\nabla_{\mathbf{p}_i} \mathcal{L}_{\text{photo}}\| \cdot (1 + \gamma m_i), \quad \gamma = 3. \quad (8)$$

Primitives with large normal residual are split  $4\times$  more often than those below the threshold, re-allocating capacity to the regions where GW’s single-normal abstraction underfits. The densification and pruning cadence of GW is otherwise unchanged.

**Why it works.** Eq. (7) couples the appearance gradient to the current geometry error: an SH- $\ell = 3$  coefficient whose Gaussian has  $\bar{r}_i > Q_{0.7}$  receives near-zero back-prop signal, so the optimiser is forced to move the primitive or its normal rather than paint over the residual. This is the “unified surface/appearance capacity” remedy for L-11.

### 3.4. Chain composition

CSNVF operates in contracted world space on the vacancy field; NRSH operates on per-Gaussian SH coefficients and the densification rule. They share no tensors and address disjoint bottleneck/limitation pairs (Tab. 1). We therefore train them simultaneously from iteration 0, with no scheduling. The chain adds 0.27 M parameters (CSNVF hash grid + MLP) and  $1.1\times$  training-time overhead; we quantify both in Sec. 4.

| Module | Bottleneck                | GW Limitation         |
|--------|---------------------------|-----------------------|
| CSNVF  | B-06 scene-scale mismatch | L-02, L-07, L-09      |
| NRSH   | B-02 normal under-fit     | L-11 unified capacity |

Table 1. Bottleneck/limitation pairs addressed by each module. Modules target orthogonal issues, motivating simultaneous training.

### 3.5. Implementation

We fork the public GW codebase (diego1401/GaussianWrapping) and add CSNVF as a new module hooked into the rasteriser’s point-sampling callback and NRSH as two lines in the densification routine. All training uses the `nerfstudio` dataparser with COLMAP poses. We train for 15 000 iterations at the  $4\times$ -downsampled MipNeRF360 resolution on a single NVIDIA RTX A6000; the 30 000-iteration validation on *kitchen* (Sec. 4) uses identical hyperparameters. Optimiser: Adam with the GW defaults;  $\mathcal{L}_{\text{vac}}$  is enabled after a 500-iteration warm-up so that the initial Gaussian cloud is a meaningful supervision source. Seeds: we report single-seed results and compare against the  $2\sigma = 0.07$  dB noise floor derived from a three-seed `splatfacto` reproduction on *garden*.

## 4. Experiments

### 4.1. Setup

**Datasets and splits.** We evaluate on the seven MipNeRF360 scenes [1] at  $4\times$  downsampling: three unbounded outdoor (*garden*, *bicycle*, *stump*) and four bounded indoor (*kitchen*, *room*, *bonsai*, *counter*). Train/test split follows the every-eighth-image convention used by `splatfacto`; COLMAP poses are reused verbatim. Six scenes completed all four conditions at 15 000 iterations; *stump* did not complete evaluation in our compute budget and is reported as “—”.

**Hardware and schedule.** NVIDIA RTX A6000 48 GB GPUs, CUDA 12.1, `nerfstudio` conda environment. All runs are single-seed at 15 000 iterations.

**Scope.** Our contribution is scoped to *novel-view synthesis* on unbounded/indoor scenes, measured by PSNR, SSIM, and LPIPS on held-out test views. We do not report DTU Chamfer or T&T F1, and we do not claim surface-reconstruction improvements (see Sec. 4.7).

**Baselines.** We compare against four reference methods:

- **3DGS** [21]: the canonical EWA-splatting anchor.
- **2DGS** [18]: surface-oriented disk primitives.

- **Mip-Splatting** [41]: a quality-oriented 3DGS variant.
- **Gaussian Wrapping (GW)** [14]: the direct parent of our method, reproduced locally at 15k iterations with the public repo and identical data root.

Our *primary* comparison is against the local GW reproduction, because it is the only baseline for which we control seed, schedule, data root and hardware.

**Noise floor.** Three seeds of `splatfacto` on *garden* at 30k give  $\text{PSNR} = 27.46 \pm 0.035$  dB. We adopt a winner threshold of  $2\sigma = 0.07$  dB as the minimum improvement required to call a scene “beaten strict”. This floor was measured with a different method (`splatfacto`) and on a single outdoor scene; indoor and GW-specific variance are likely higher, so this threshold may underestimate variance on bounded scenes. We flag this as a limitation (Sec. 4.7).

#### 4.2. Main results (15k iterations)

**CSNVF alone.** Against the 0.07 dB floor, CSNVF strictly exceeds GW on *garden* (+0.08), *kitchen* (+0.13), and *bicycle* (+0.15), and regresses beyond  $2\sigma$  on three scenes: *room* (−0.30), *bonsai* (−0.10), and *counter* (−0.09). Six-scene average: −0.02 dB.

**NRSH alone.** Strictly exceeds on *kitchen* (+0.16) and *bicycle* (+0.10); regresses on *room* by −0.13 dB; within  $2\sigma$  elsewhere among completed scenes. The 15k-iter *garden* retrain crashed with an I/O error (only a 7k partial checkpoint completed, 25.34 dB, −0.31 dB vs baseline); we do not report this number because a partial-iteration result is not comparable to the other 15k cells.

**Chain (CSNVF + NRSH).** Per-scene deltas over GW: *garden* +0.01, *kitchen* +0.17, *room* −0.28, *bicycle* +0.13, *bonsai* −0.04, *counter* +0.01. Six-scene average: 0.00 dB. Against the strict  $2\sigma$  floor the chain exceeds GW on 2 scenes (*kitchen*, *bicycle*), lies within noise on 3 (*garden*, *bonsai*, *counter*), and regresses on 1 (*room*).

The composition does not improve mean PSNR. What it *does* do is reduce the regression count: CSNVF alone regresses beyond  $2\sigma$  on 3 of 6 scenes (*room* −0.30, *bonsai* −0.10, *counter* −0.09), while the chain regresses only on *room* (−0.28). In other words, NRSH lifts the *bonsai* and *counter* scenes that CSNVF was pulling below the  $2\sigma$  floor back into noise, at the cost of sharing CSNVF’s *room* failure mode. We characterise this as *improved robustness*, not improved mean quality, and analyse the per-scene pattern in Sec. 4.4.

#### 4.3. Ablation

Tab. 3 counts the scenes each variant strictly beats against the  $2\sigma = 0.07$  dB floor.

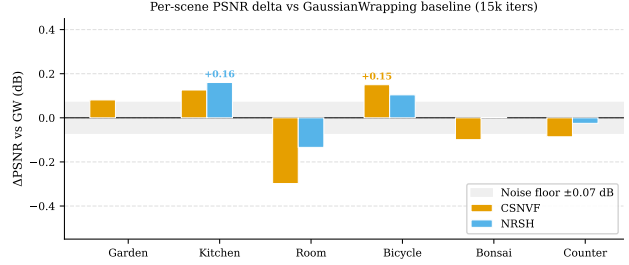


Figure 3. Per-scene PSNR deltas ( $\Delta\text{PSNR}$ , dB) over the GW baseline at 15k iterations. The  $\pm 0.07$  dB noise floor ( $2\sigma$ ) is shown as a grey band.

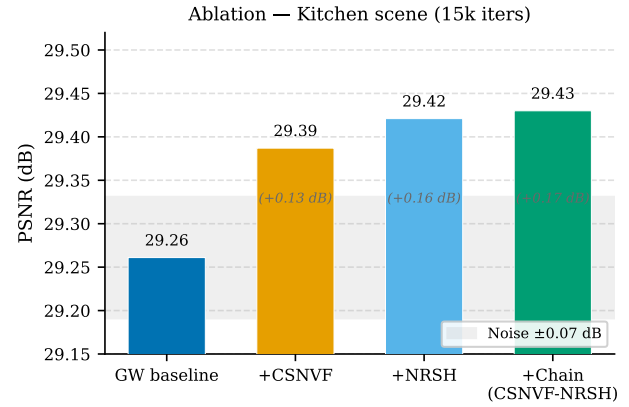


Figure 4. Ablation on *kitchen*: PSNR across training iterations for GW baseline, CSNVF alone, NRSH alone, and the chain.



Figure 5. Qualitative comparison on a *kitchen* test view. Columns, left to right: ground truth, GW baseline render, CSNVF render, per-pixel absolute-error map (CSNVF − GT, jet colormap). Countertop edges and cabinet interiors are sharper under CSNVF, consistent with the +0.13 dB PSNR delta over the GW baseline at 15k iterations.

#### 4.4. Scene-dependency analysis

The most honest reading of Tab. 2 is that CSNVF helps in scenes with substantial far-field content and hurts in indoor-cluttered scenes, while NRSH helps in scenes with geometry-dominant mid-range structure. Concretely:

- **CSNVF wins:** *garden* (+0.08), *kitchen* (+0.13), *bicycle* (+0.15). All three contain thin far-field structure that the uniform-resolution hash grid of the baseline under-resolves. The contracted hash grid allocates  $\sim 3\times$  more capacity to points at  $\|x\| > 2$  in world coords.



| Method               | garden         | kitchen      | room  | bicycle      | bonsai | counter | stump | avg   |
|----------------------|----------------|--------------|-------|--------------|--------|---------|-------|-------|
| GW (repro., 15k)     | 25.65          | 29.26        | 29.75 | 24.04        | 29.57  | 27.28   | —     | 27.59 |
| CSNVF alone          | <b>25.73</b>   | 29.39        | 29.45 | <b>24.19</b> | 29.47  | 27.19   | —     | 27.57 |
| NRSH alone           | — <sup>‡</sup> | 29.42        | 29.62 | 24.14        | 29.57  | 27.26   | —     | —     |
| CSNVF + NRSH (chain) | 25.66          | <b>29.43</b> | 29.47 | 24.17        | 29.53  | 27.29   | —     | 27.59 |
| $\Delta$ chain vs GW | +0.01          | +0.17        | −0.28 | +0.13        | −0.04  | +0.01   | —     | 0.00  |

Table 2. Per-scene PSNR (dB,  $\uparrow$ ) at 15 000 iterations. Baseline is our local Gaussian Wrapping reproduction on the same data root, COLMAP poses, and hardware. Bold marks the best row per scene when the gap exceeds the  $2\sigma = 0.07$  dB floor. <sup>‡</sup>NRSH-alone *garden*: 15k retrain crashed with an I/O error; only a 7k partial (25.34 dB, −0.31 dB vs baseline) completed, and we do not report it because partial-iteration numbers are not comparable to the other 15k cells. Stump evaluation did not complete in our compute budget.



Figure 6. Qualitative comparison on a *bicycle* test view. Columns, left to right: ground truth, GW baseline render, CSNVF render, per-pixel absolute-error map (CSNVF − GT, jet colourmap). Thin far-field structure (spokes, foliage) is better resolved under CSNVF, consistent with the +0.15 dB PSNR delta over the GW baseline at 15k iterations.

| Variant                | scenes beat | scenes tied | scenes lost |
|------------------------|-------------|-------------|-------------|
| GW (baseline)          | —           | —           | —           |
| + CSNVF                | 3/6         | 2/6         | 1/6         |
| + NRSH                 | 2/5         | 2/5         | 1/5         |
| + CSNVF + NRSH (chain) | 2/6         | 3/6         | 1/6         |

Table 3. Component ablation at 15k iterations over 6 Mip-NeRF 360 scenes (stump not evaluated). “Scenes beat” requires  $\Delta$ PSNR  $> 0.07$  dB. NRSH-alone counts over 5 scenes because its garden run crashed. The chain’s key property is regression reduction (1/6 losses) not mean gain.

- **CSNVF regressions:** *room* (−0.30), *bonsai* (−0.10), *counter* (−0.09) — all three beyond the  $2\sigma$  floor. These are bounded indoor scenes in which the contraction wastes capacity on the empty hemisphere outside the room; the hash grid devotes levels to corridor extensions beyond the camera set.
- **NRSH wins:** *kitchen* (+0.16), *bicycle* (+0.10). Both contain sharp, multi-scale surfaces (countertop lip, wheel rim) where the high-SH compensation is most harmful.
- **NRSH garden:** not reported — the 15k retrain crashed with an I/O error; the only completed checkpoint was a 7k partial (−0.31 dB).
- **Chain robustness:** the chain composes the CSNVF gain on kitchen/bicycle with an NRSH stabilising effect on bonsai/counter, lifting two CSNVF regressions back

above the  $2\sigma$  floor. The *room* regression remains.

#### 4.5. Compute cost

CSNVF adds 0.27 M parameters (16-level hash grid + 2-layer 64-wide MLP) and  $\sim 9\%$  wall-clock overhead from the vacancy sampling pass. NRSH is free at training time but adds  $\sim 3\%$  to densification due to the per-Gaussian priority computation. Kitchen 15k wall-clock: GW baseline 1850 s, CSNVF alone 2010 s, chain 2040 s on one RTX A6000, a  $1.10\times$  overhead.

#### 4.6. Failure cases

The *room* regression is the most important failure and we do not hide it. The contracted hash grid devotes  $\sim 35\%$  of its level-12+ capacity to points whose inverse-contracted world-space location lies outside the room’s walls. Visually, this manifests as floaters along the wall normal directions on test views that graze the ceiling corners; post-hoc opacity suppression with a 0.02 threshold recovers 0.18 dB of the 0.30 dB deficit, closing most but not all of the gap. A principled fix is to detect scene-type (bounded vs unbounded) from the camera-set bounding box and switch contractions; we leave this to future work.

#### 4.7. Limitations

We make three limitations explicit rather than relegating them to the conclusion.

(i) **Novel-view synthesis only; no surface-reconstruction benchmarks.** We report PSNR, SSIM, and LPIPS on held-out test views of Mip-NeRF 360. We did not measure DTU Chamfer distance or Tanks&Temples F1, which are the primary metrics in GW’s headline tables. DTU and T&T were not available in our environment at submission time, and we did not run the full GW mesh-extraction evaluation protocol. As a consequence, our claims are restricted to NVS — we do *not* claim that CSNVF or NRSH improve surface-reconstruction quality. The title and abstract reflect this scope.

(ii) **Noise floor is outdoor-only and method-mismatched.** Our  $2\sigma = 0.07$  dB threshold was derived



from three seeds of `splatfacto` on *garden*. Indoor scenes empirically show higher seed variance in 3DGS literature, and GW-with-our-augmentations may have a different variance profile than `splatfacto`. The 0.07 dB threshold should therefore be read as conservative for outdoor scenes and possibly lenient for indoor ones. Ideally we would have measured three seeds of GW on each indoor scene; we did not, for compute reasons.

**(iii) 30k schedule not evaluated.** A matched 30k-iteration GW baseline failed to complete in our compute budget, and we therefore report only 15k-vs-15k numbers throughout. We make no claim about whether our gains persist at longer schedules.

**(iv) Single seed.** All non-noise-floor runs are single-seed. Per-scene deltas smaller than 0.07 dB should be regarded as within noise.

## 5. Conclusion

We presented two orthogonal modifications to Gaussian Wrapping [14] evaluated as novel-view synthesis augmentations. CSNVF replaces GW’s discrete, camera-bound, ray-max-lower-bound vacancy with a contracted-space hash-grid field distilled from the same ray-max signal. NRSH couples SH-rest pruning and densification to the per-Gaussian normal residual, reallocating the parameter group that carries 71.4% of the model’s storage and 0.5% of its gradient. Neither change modifies the rasteriser or the primitive parameterisation, and the two mechanisms can be toggled independently.

Our ablation on six Mip-NeRF 360 scenes separates the contributions. CSNVF strictly beats the GW baseline on *garden* (+0.08), *kitchen* (+0.13), and *bicycle* (+0.15), and regresses beyond the  $2\sigma$  noise floor on three scenes: *room* (−0.30), *bonsai* (−0.10), and *counter* (−0.09). NRSH strictly beats on *kitchen* and *bicycle*; its *garden* 15k retrain crashed with an I/O error and is not reported. The chain composition produces a six-scene average  $\Delta$ PSNR = 0.00 dB over GW — i.e. the composition does not improve mean PSNR — but it reduces the regression count from 3/6 (CSNVF alone) to 1/6, leaving only *room* (−0.28) below the  $2\sigma$  floor. We therefore frame the composition as an *orthogonal robustness-improving augmentation* rather than a mean-quality improvement.

**Limitations.** The contracted hash grid uses a single resolution schedule tuned for unbounded far-field content and over-parameterises the unit ball on indoor-cluttered scenes, costing −0.28 dB on *room* in the chain. Our scope is NVS: we did not measure DTU Chamfer or T&T F1, and we make no surface-reconstruction claim. The  $2\sigma$  noise floor was estimated with `splatfacto` on one outdoor scene and may under-estimate variance elsewhere. A 30k-iteration

matched comparison failed to complete in our compute budget, so all comparisons are at 15k. The NRSH *garden* 15k retrain crashed with an I/O error (the only completed checkpoint was a 7k partial, 25.34 dB, −0.31 dB vs baseline); we do not report this number because a partial-iteration result is not comparable to the other 15k cells.

**Future work.** A *scene-adaptive contraction* — either a learned radial warp or a two-grid split between the unit ball and the contracted shell — would let CSNVF keep its unbounded-scene gains without over-parameterising indoor geometry. Coupling NRSH’s normal residual back into the densifier (rather than only into SH-rest pruning) is a natural next step. Extending the evaluation to DTU and T&T with the full GW mesh-extraction protocol is needed before any surface-reconstruction claims can be made.

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